TUNIS BUSINESS SCHOOL
UNIVERSITY OF TUNIS

University of Tunis
Tunis Business School **Fall 2022**

Production & Operation Management BA 305
Midterm Exam
(Duration 1.5 hours)

Question 1

An agricultural Co. has 600 hectares (he) of land. The annual cost to exploit this land is 0.5 million TND. The land yields 1.5 ton of wheat per he. One ton of wheat generates 750 TND.

1. Provide the current productivity of land exploitation of the Co.
2. An improvement plan consists of introducing a computerized irrigation system, qualified labor, an optimized system of fertilizers' allocation. The total annual additional costs are respectively of 200,000 TND, 100,000 TND and 50,000 TND, respectively. The new land productivity is of 3.5 ton/he. Identify the new company productivity of land exploitation posterior to the implemented improvements.

Question 2 Exercice incorrect

We consider a house-building project consisting of seven activities. The precedence orders, durations, and costs are given in the table below:

Activity	predecessor	Regular time (weeks)	Crash time (weeks)	Normal cost	Crash cost
1	-	12	7	3000	5000
2	1	8	5	2000	3500
3	2	4	3	4000	7000
4	2	4	1	500	1100
5	4,3	12	9	50000	71000
6	4,5,	4	1	500	1100
7	6	4	3	15000	22000

1- Calculate the minimum project's completion time.
2- What is the cost of the project?
3- The house owner wants to reduce the completion time to 25 weeks. Provide the optimal project crashing plan and its corresponding cost.

Scanné avec CamScanner

Question 3

The demand for computer services at PM company over the last 12 weeks is given by

Period	Demand
1	37
2	40
3	41
4	37
5	45
6	50
7	43
8	47
9	56
10	52
11	55
12	54

1- Use a three-week moving average from week 4 to week 12 and calculate the mean squared error.
2- Repeat question 1 with weighted moving average with weights of 4, 2, and 1 respectively.
3- Provide the computational details for the exponential smoothing forecast with $\alpha=0.3$ assuming that the forecast for period 1 was 39 and evaluate the corresponding mean absolute deviation.

Scanne avec CamScanner